AZIM PREMJI UNIVERSITY, BHOPAL
First Semester 2026–2027
Tutorial Sheet–9

Course No. MTH-113 Course title: Introduction to Quantitative Thinking

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(1) The blood groups of 30 students of Class IX are recorded as follows:

$$A, B, O, O, AB, O, A, O, B, A, O, B, A, O, O,$$
$$A, AB, O, A, A, O, O, AB, B, A, O, B, A, B, O.$$

Represent this data in the form of a frequency distribution table. Which is the most common, and which is the rarest, blood group among these students?

(2) The heights of 50 students, measured to the nearest centimetres, have been found to be as follows:

161	150	154	165	168	161	154	162	150	151
162	164	171	165	158	154	156	172	160	170
153	159	161	170	162	165	166	168	165	164
154	152	153	156	158	162	160	161	173	166
161	159	162	167	168	159	158	153	154	159

(a) Represent the data given above by a grouped frequency distribution table, taking the class intervals as 160 - 165, 165 - 170, etc.
(b) What can you conclude about their heights from the table?

(3) Three coins were tossed 30 times simultaneously. Each time the number of heads occurring was noted down as follows:

0	1	2	2	1	2	3	1	3	0
1	3	1	1	2	2	0	1	2	1
3	0	0	1	1	2	3	2	2	0

Prepare a frequency distribution table for the data given above.

(4) The value of π upto 50 decimal places is given below:

3.14159265358979323846264338327950288419716939937510

(a) Make a frequency distribution of the digits from 0 to 9 after the decimal point.
(b) What are the most and the least frequently occurring digits?

(5) A survey conducted by an organisation for the cause of illness and death among the women between the ages 15 - 44 (in years) worldwide, found the following figures (in %):

S.No.	**Causes**	**Female fatality rate (%)**
1.	Reproductive health conditions	31.8
2.	Neuropsychiatric conditions	25.4
3.	Injuries	12.4
4.	Cardiovascular conditions	4.3
5.	Respiratory conditions	4.1
6.	Other causes	22.0

(a) Represent the information given above graphically.
(b) Which condition is the major cause of women's ill health and death worldwide?
(c) Try to find out, with the help of your teacher, any two factors which play a major role in the cause in (b) above being the major cause.

(6) The following data on the number of girls (to the nearest ten) per thousand boys in different sections of Indian society is given below.

Section	**Number of girls per thousand boys**
Scheduled Caste (SC)	940
Scheduled Tribe (ST)	970
Non SC/ST	920
Backward districts	950
Non-backward districts	920
Rural	930
Urban	910

(a) Represent the information above by a bar graph.
(b) In the classroom discuss what conclusions can be arrived at from the graph.

(7) The length of 40 leaves of a plant are measured correct to one millimetre, and the obtained data is represented in the following table:

Length (in mm)	**Number of leaves**
118 - 126	3
127 - 135	5
136 - 144	9
145 - 153	12
154 - 162	5
163 - 171	4
172 - 180	2

(a) Draw a histogram to represent the given data.
(b) Is there any other suitable graphical representation for the same data?
(c) Is it correct to conclude that the maximum number of leaves are 153 mm long? Why?

(8) The following table gives the distribution of students of two sections according to the marks obtained by them:

Section A		**Section B**	
Marks	**Frequency**	**Marks**	**Frequency**
0 - 10	3	0 - 10	5
10 - 20	9	10 - 20	19
20 - 30	17	20 - 30	15
30 - 40	12	30 - 40	10
40 - 50	9	40 - 50	1

Represent the marks of the students of both the sections on the same graph by two frequency polygons. From the two polygons compare the performance of the two sections.

(9) The runs scored by two teams A and B on the first 60 balls in a cricket match are given below:

Number of balls	**Team A**	**Team B**
1 - 6	2	5
7 - 12	1	6
13 - 18	8	2
19 - 24	9	10
25 - 30	4	5
31 - 36	5	6
37 - 42	6	3
43 - 48	10	4
49 - 54	6	8
55 - 60	2	10

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Represent the data of both the teams on the same graph by frequency polygons. [**Hint:** First make the class intervals continuous.]

(10) The following number of goals were scored by a team in a series of 10 matches:

$$2, \quad 3, \quad 4, \quad 5, \quad 0, \quad 1, \quad 3, \quad 3, \quad 4, \quad 3$$

Find the mean, median and mode of these scores.

(11) In a mathematics test given to 15 students, the following marks (out of 100) are recorded:

$$41, \quad 39, \quad 48, \quad 52, \quad 46, \quad 62, \quad 54, \quad 40, \quad 96, \quad 52, \quad 98, \quad 40, \quad 42, \quad 52, \quad 60$$

Find the mean, median and mode of this data.

(12) The following observations have been arranged in ascending order. If the median of the data is 63, find the value of x.

$$29, \quad 32, \quad 48, \quad 50, \quad x, \quad x+2, \quad 72, \quad 78, \quad 84, \quad 95$$

(13) Find the mean salary of 60 workers of a factory from the following table:

Salary (in Rs)	**Number of workers**
3000	16
4000	12
5000	10
6000	8
7000	6
8000	4
9000	3
10000	1
Total	60

(14) Give one example of a situation in which
  (a) the mean is an appropriate measure of central tendency.
  (b) the mean is not an appropriate measure of central tendency but the median is an appropriate measure of central tendency.

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